I

□□□□□□□□□□□□□□□

20 2026

2023 2026

2026 9

- 70% 70%
-
-
-
- SMAPE

1

2025 2026 0.70

□□	□□□□
	2025
	0.70

70%

2

0 1

$$E = \frac{\ln(1+n)}{\ln(1+n)}$$

n_o o n_{max} 1 0

□□□□□	□□□□□□□□
	2%
	5%
	20

□□□□□	□□□□□□□□
	4%

6

3

58%

$$S = 0.38S + 0.17S + 0.14S + 0.08I + 0.08I + 0.08S + 0.07S$$

F1

$$\text{Precision} = \frac{M}{N} \quad \text{Recall} = \frac{M}{N} \quad \text{F1} = \frac{2\text{Precision}\text{Recall}}{\text{Precision} + \text{Recall}}$$

M N_p N_a F1

□□	□□□□□□	□□
	55.77%	826 ÷ 1,481
	52.68%	826 ÷ 1,568
F1	54.18%	

$$S = 0.50\text{F1} + 0.20S + 0.15A + 0.10A + 0.05A$$

□□	□□	□□
F1	50%	
	20%	
	15%	1 SMAPE 0
	10%	180 0
	5%	

$$A = \text{mean}\left[\max\left(0, 1 - \frac{|\Delta\text{days}|}{180}\right)\right] \quad A = \max\left(0, 1 - \frac{|N - N|}{N}\right)$$

4

$$S = 0.46P + 0.20P + 0.09P + 0.08H + 0.06H + 0.05T + 0.04T + 0.02E$$

□□	□□	□□
2025	46%	
2024	20%	
2023	9%	
2025	8%	
2025	6%	
	5%	
	4%	
	2%	

$$N = \text{round}\left(0.60N + 0.30N + 0.10N\right)$$

1,069 1,069 60% 30% 10%
EWMA

5

1 2 3 4

□□	□□□	□□
	70	70
	5	5
	12	12

□□	□□□	□□
	4	20 √20
	0 1	0 1

$$\mathrm{RF}\left(X\right) = \frac{1}{70}\sum_{k=1}^{70}\mathrm{Tree}\left(X\right)$$

70

6

$$r = y - \hat{y} \quad \hat{y} = \hat{y} + 0.06\mathrm{Tree}\left(X\right)$$

□□	□□□	□□
	55	55
	0.06	6%
	2	
	15	15

1 0.4 +0.6 0 0.7 −0.7 0.06

7

$$S = 0.22R\left(\mathrm{EWMA}\right) + 0.18\mathrm{Trend} + 0.25R\left(\mathrm{RF}\right) + 0.25R\left(\mathrm{GBT}\right) + 0.06R\left(T\right) + 0.04E$$

R 0 1

□□	□□	□□□□
	22%	
	18%	
	25%	
	25%	

□□	□□	□□□□
	6%	
	4%	

$$\text{Trend} = \max\left(0, C + 0.25\left(C - C\right)\right)$$

25% 2025 90% 110% 50% 150%

22% 18% 25% 25% 6% 4% AI
50% 40% 10%

2024 2025 2026

8

2026 e N(y,e) e y ln(1+x)

□□	□□□□	□□□□	□□□□
1	count_lag1	2025	N(2025,e)
2	count_lag2	2024	N(2024,e)
3	count_lag3	2023	N(2023,e)
4	present_lag1	2025	1 0
5	present_lag2	2024	1 0
6	present_lag3	2023	1 0
7	recency		2025 1 2024 1/2 2023 1/3 0
8	item_demand_lag1	2025	ln(1+2025)
9	item_demand_lag2	2024	ln(1+2024)
10	org_demand_lag1	2025	ln(1+2025)
11	org_demand_lag2	2024	ln(1+2024)
12	region_demand_lag1	2025	ln(1+2025)
13	amount_lag1	2025	ln(1+ 2025) 0
14	amount_lag2	2024	ln(1+ 2024) 0

□□	□□□□	□□□□	□□□□
15	amount_change		amount_lag1 amount_lag2
16	start_month		2025 ÷30.5÷12 0.5
17	duration		2025 0 730 ÷365 0.12
18	transaction_lag1		ln(1+2025)
19	transaction_entity_lag1		ln(1+2025)
20	project_exposure		ln(1+)÷ln(1+)

7 6 6 1 7 7 5 1 20

9

P A

$$S = \max\left(0, 1 - \frac{\left|\ln(1+P) - \ln(1+A)\right|}{\ln(5)}\right)$$

800 900

$$1 - \frac{\left|\ln(801) - \ln(901)\right|}{\ln(5)} = 0.926903 \approx 92.7\%$$

92.7% ln5 5 0 1 0

$$\text{SMAPE} = \frac{|P - A|}{(|P| + |A|) / 2} \qquad A = \max\left(0, 1 - \min\left(\bar{\text{SMAPE}}, 1\right)\right)$$

800 900 SMAPE 100÷850 11.76% 88.24% 92.7% 88.24%
7% 15%

□□□□□□□□□□	□□□□
70% 70%	70%
AI	

□□□□□□□□□□	□□□□
	F1
1 2	

2026